

CLOUD COMPUTING LAB

Experiment 1

Installation of Hypervisors and Initiation of Virtual Machines with Image File

Student: Abhijit Das

Virtualization Platform: Oracle VirtualBox

Guest Operating System: Ubuntu 25.04 (64-bit)

Virtual Machine: Ubuntu-Cloud-Lab / Ubuntu-Cloud-Lab Clone

Objective

Install and configure an open-source hypervisor, create and run a virtual machine using an OS image, configure its hardware and network, and install the required build tools and Linux kernel headers.

Result: The Ubuntu virtual machine is running successfully, the required build tools and kernel headers have been installed, and the environment is verified using GCC, Make, and the running kernel version.

1. Oracle VirtualBox Manager — VM Configuration

Shows the Ubuntu-Cloud-Lab VM with its current operating-system, CPU, RAM, storage and NAT configuration.

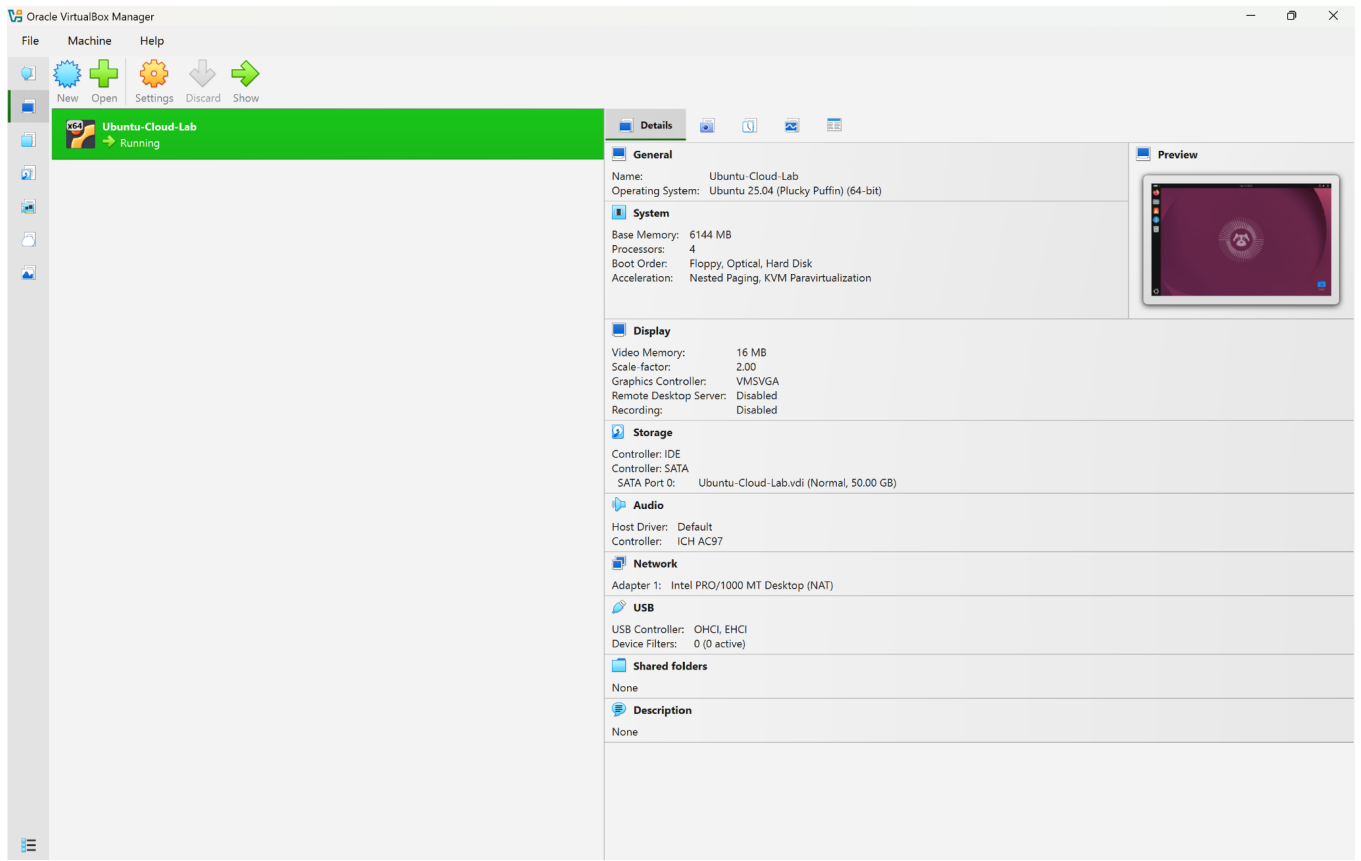


Figure 1. 1. Oracle VirtualBox Manager — VM Configuration

2. Storage Configuration

Shows the VM storage controller and the virtual disk attached to the SATA controller.

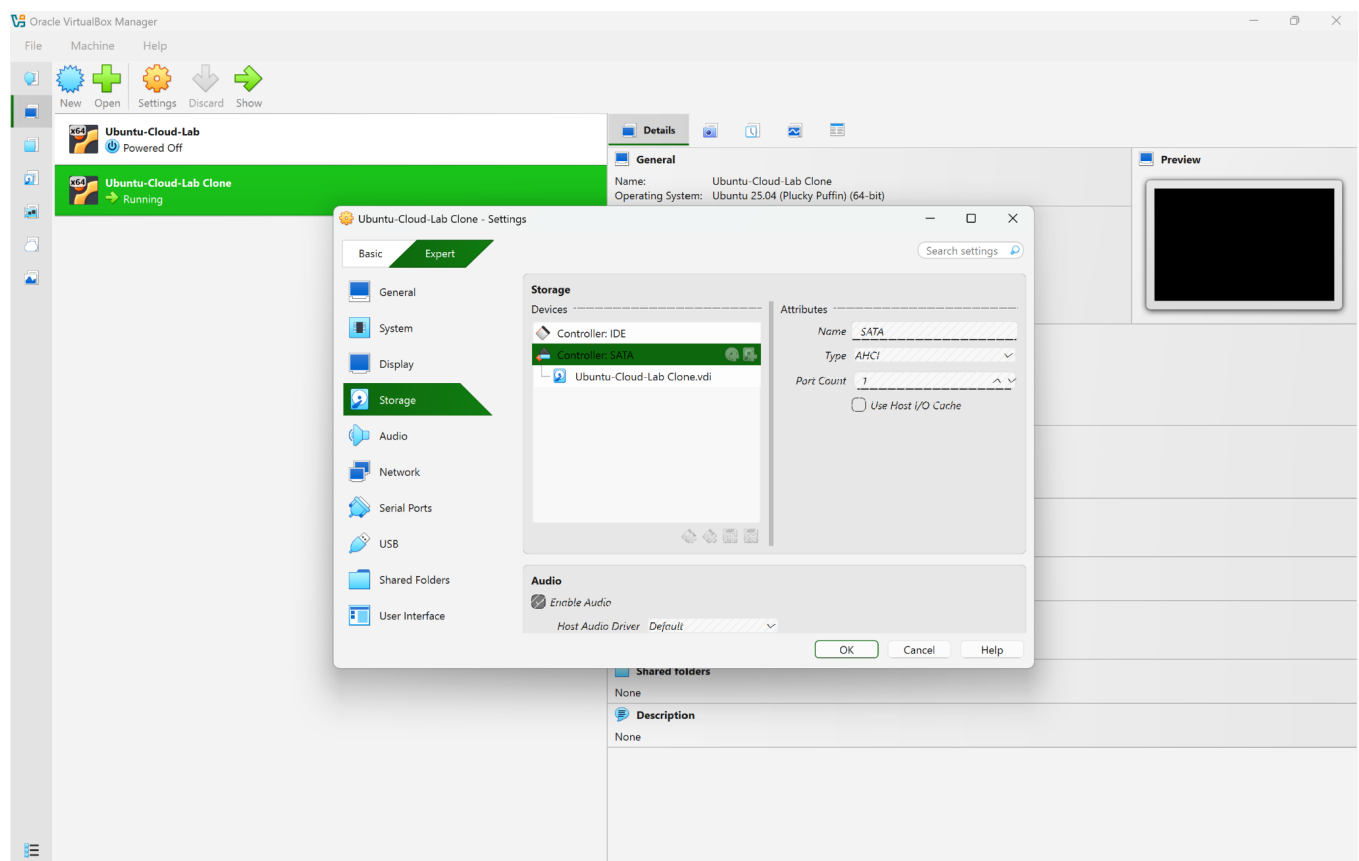


Figure 2. 2. Storage Configuration

3. Network Configuration

Shows Adapter 1 enabled and connected to the CloudLab-Network NAT Network.

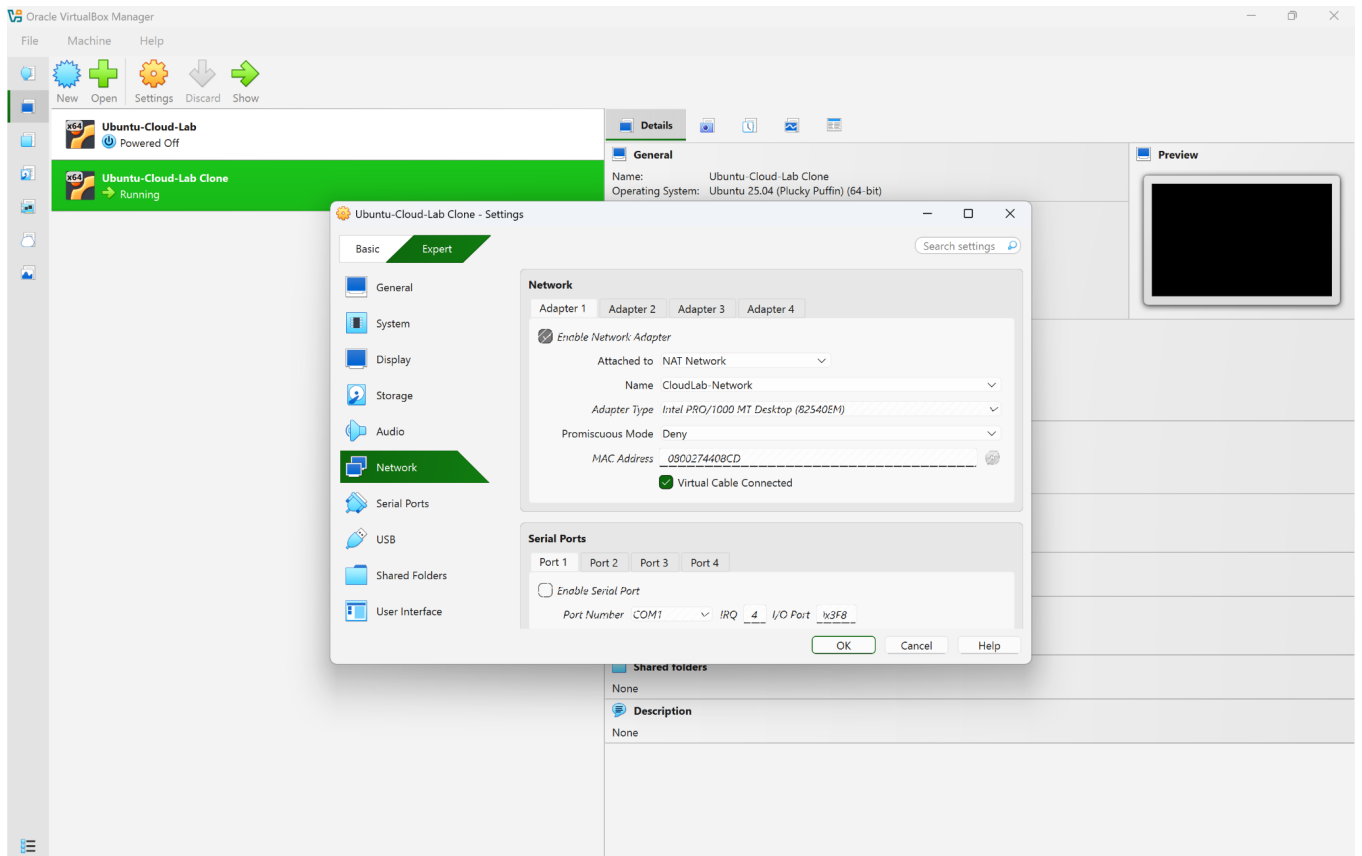


Figure 3. 3. Network Configuration

4. Ubuntu User Login

Shows the Ubuntu login screen for the configured Abhijit Das user.

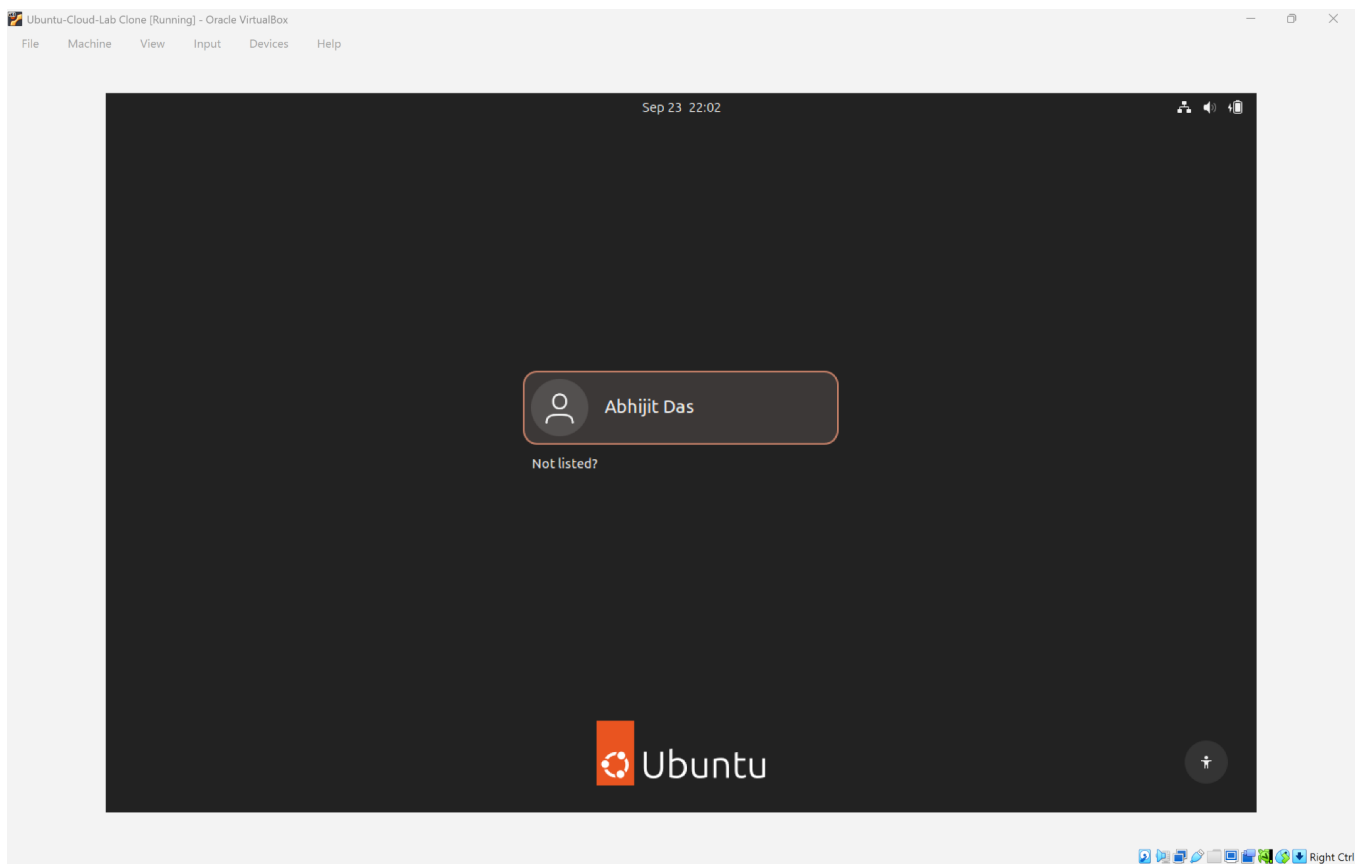


Figure 4. 4. Ubuntu User Login

5. Ubuntu Home Screen — VM Running

Shows the successfully booted Ubuntu desktop inside the VirtualBox VM.

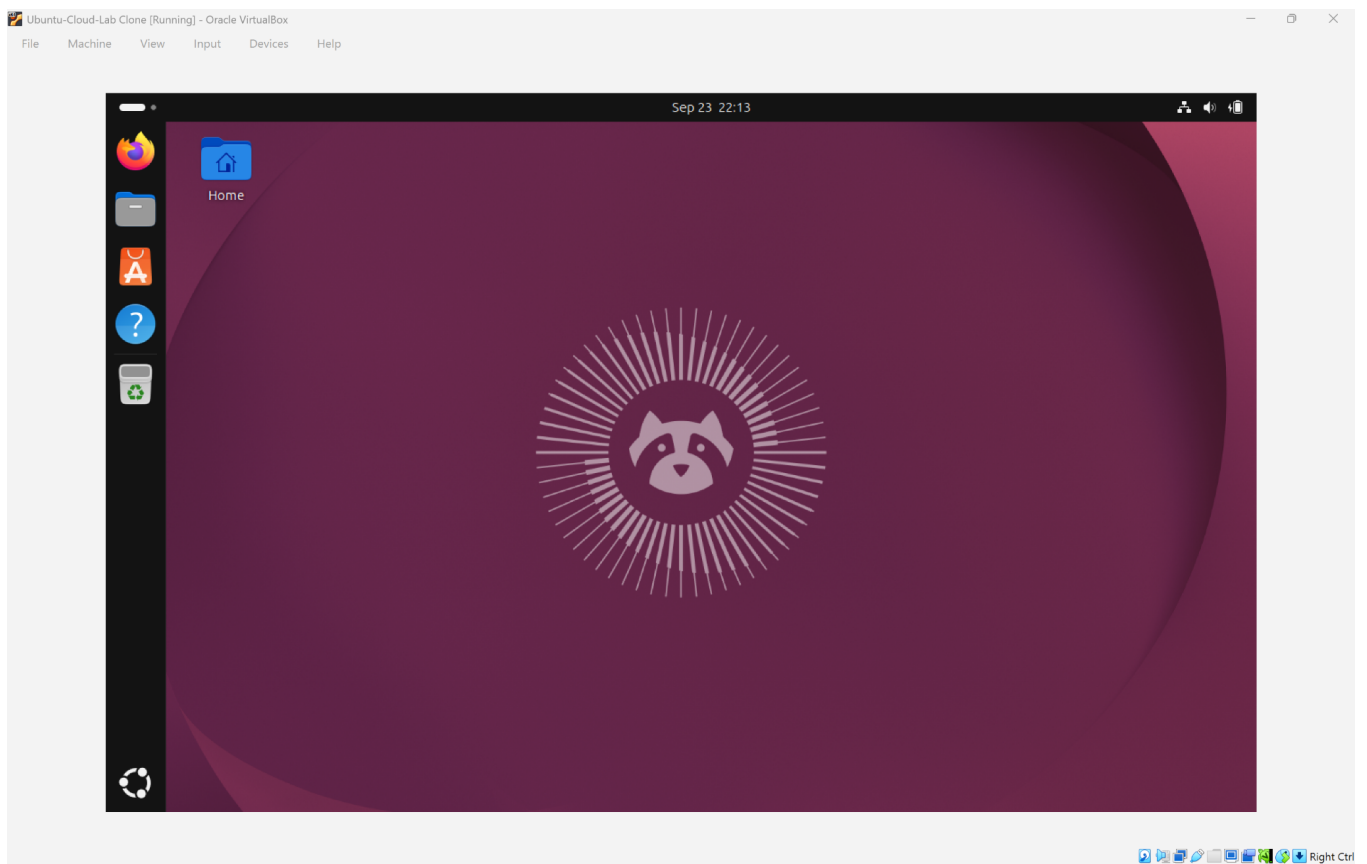
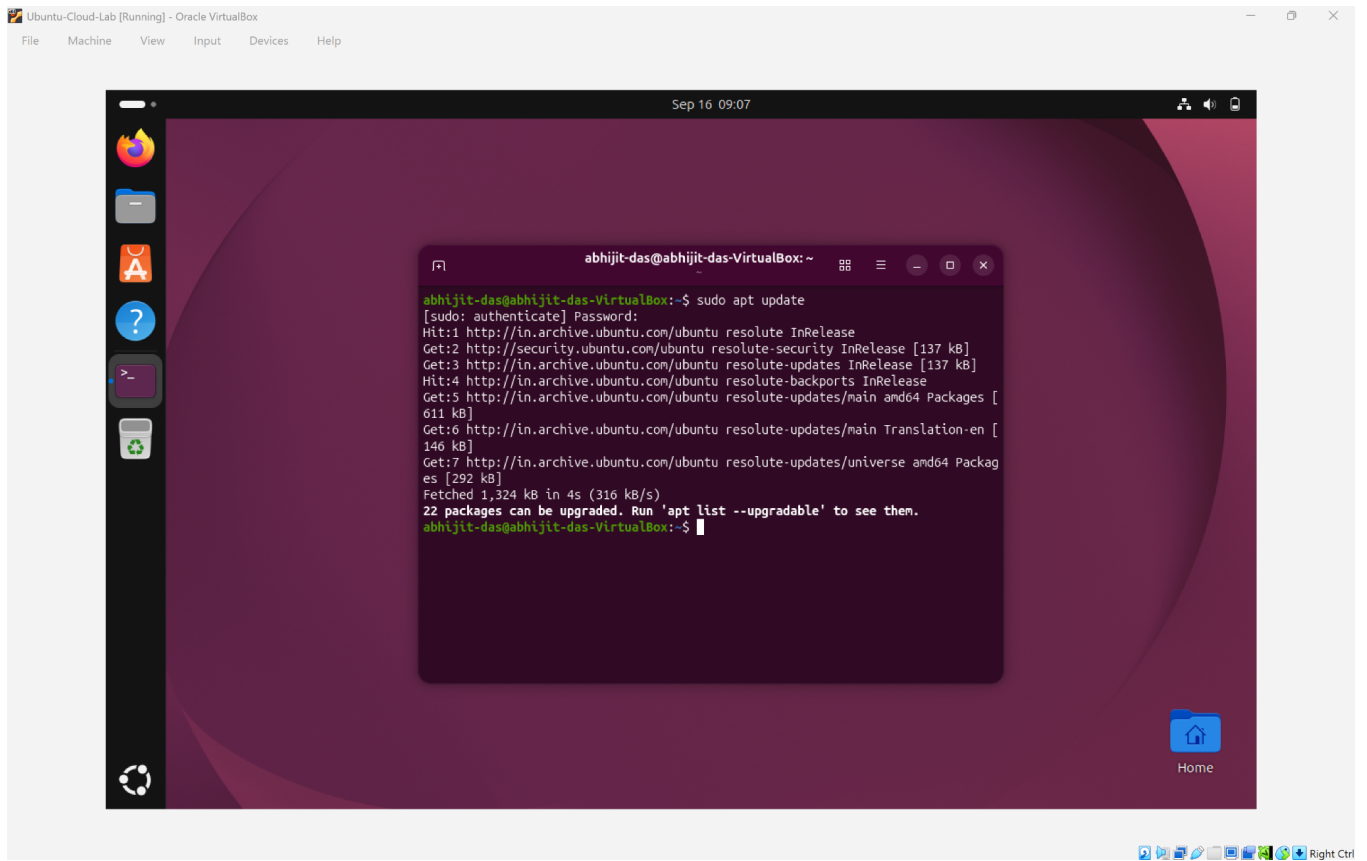


Figure 5. 5. Ubuntu Home Screen — VM Running

6. Updating Ubuntu Package Lists

Shows successful execution of `sudo apt update` inside the virtual machine.

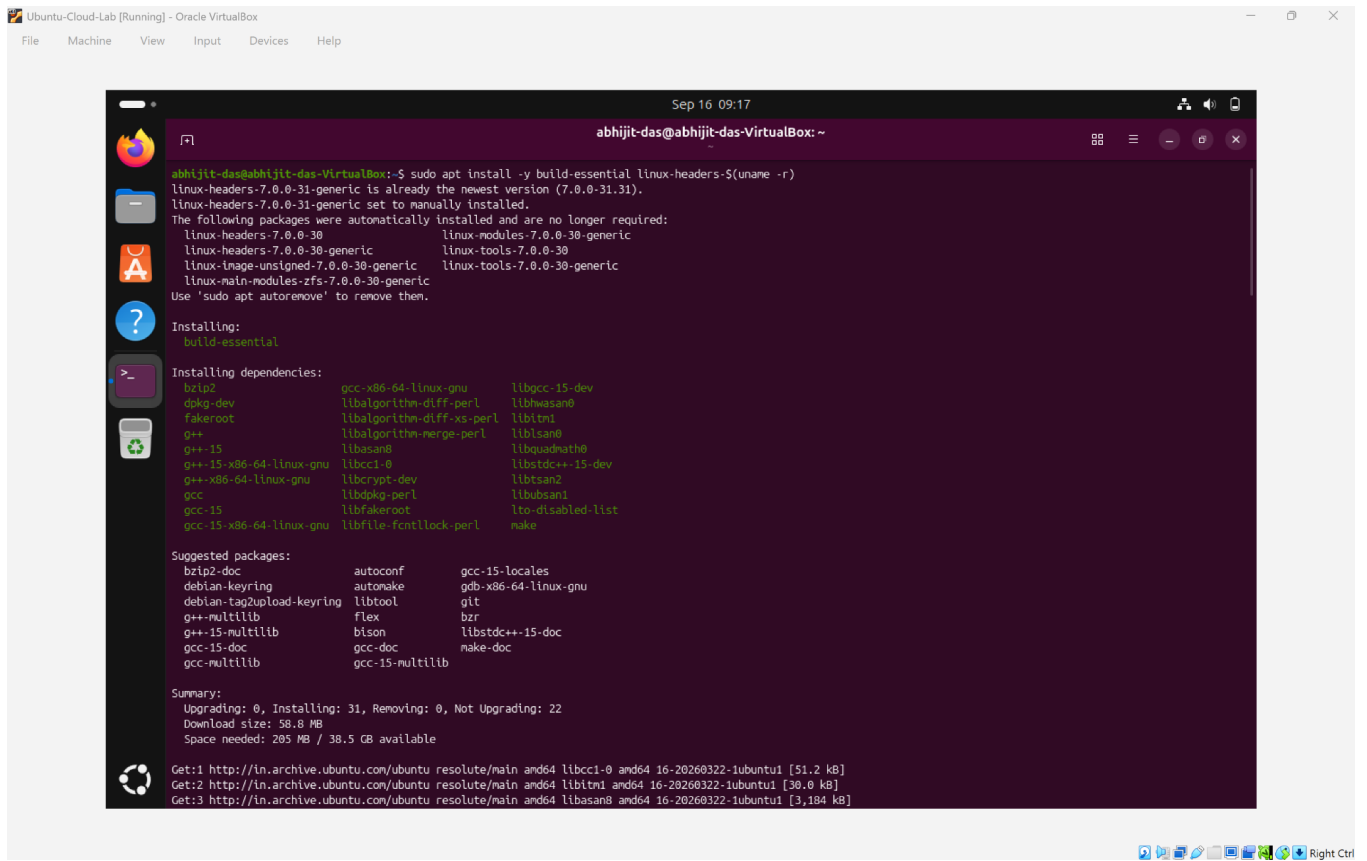


```
abhihit-das@abhihit-das-VirtualBox: ~  
abhihit-das@abhihit-das-VirtualBox:~$ sudo apt update  
[sudo: authenticate] Password:  
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease  
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease  
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [611 kB]  
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [146 kB]  
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [292 kB]  
Fetched 1,324 kB in 4s (316 kB/s)  
22 packages can be upgraded. Run 'apt list --upgradable' to see them.  
abhihit-das@abhihit-das-VirtualBox:~$
```

Figure 6. 6. Updating Ubuntu Package Lists

7. Installing Build Tools and Kernel Headers

Shows execution of `sudo apt install -y build-essential linux-headers-$(uname -r)` and the packages being installed.



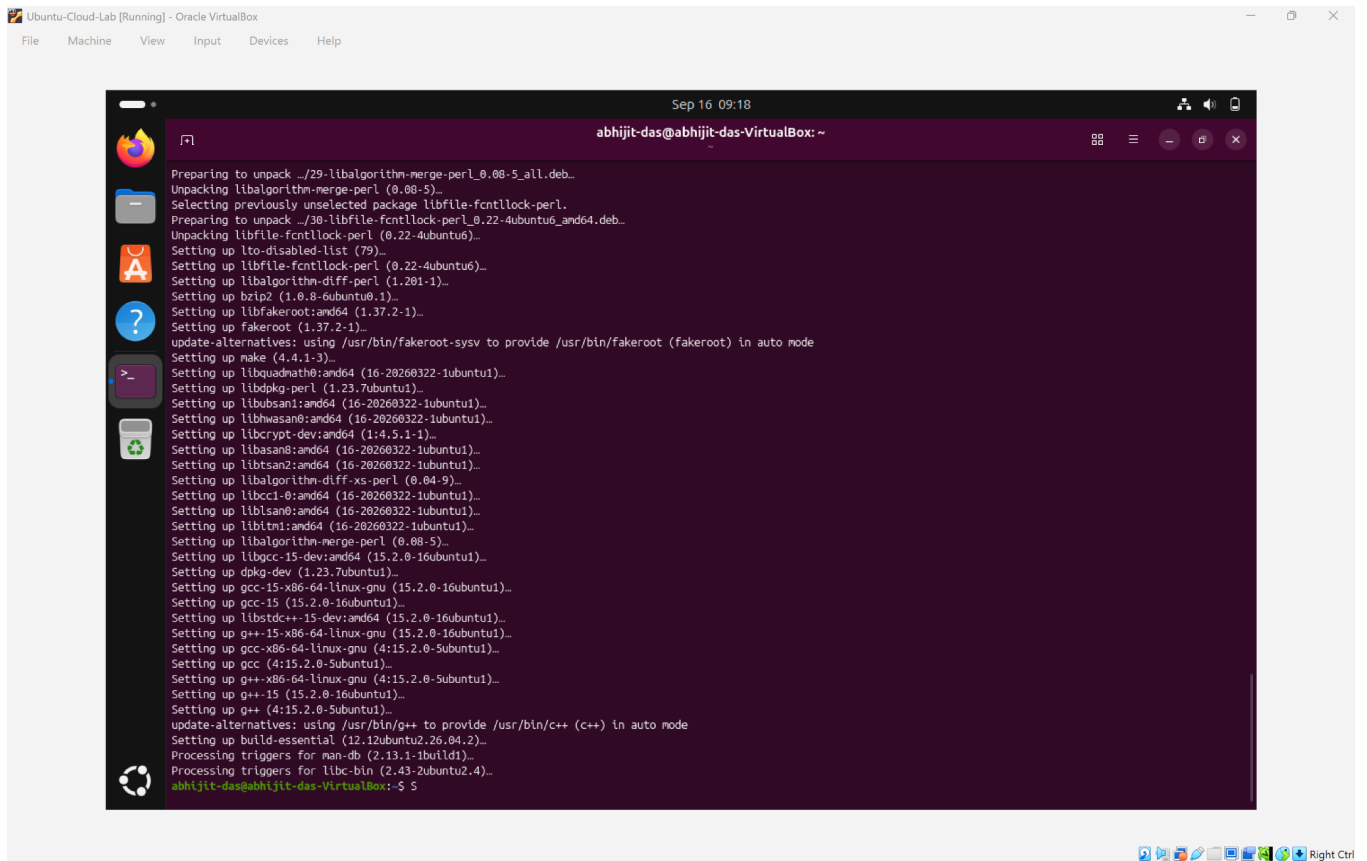
The screenshot shows a terminal window titled "abhihit-das@abhihit-das-VirtualBox: ~" with a timestamp of "Sep 16 09:17". The terminal displays the output of the command `sudo apt install -y build-essential linux-headers-$(uname -r)`. The output indicates that `linux-headers-7.0.0-31-generic` is already installed. It then lists several packages that will be installed: `linux-headers-7.0.0-30`, `linux-headers-7.0.0-30-generic`, `linux-image-unsigned-7.0.0-30-generic`, `linux-main-modules-zfs-7.0.0-30-generic`, `linux-modules-7.0.0-30-generic`, `linux-tools-7.0.0-30`, and `linux-tools-7.0.0-30-generic`. The terminal also shows the installation of `build-essential` and its dependencies, including `gcc-x86-64-linux-gnu`, `libgcc-15-dev`, `dpkg-dev`, `libalgorithm-diff-perl`, `fakeroot`, `libalgorithm-diff-xs-perl`, `g++`, `libalgorithm-merge-perl`, `g++-15`, `libasan8`, `libquadmath0`, `g++-15-x86-64-linux-gnu`, `libcc1-0`, `libstdc++-15-dev`, `g++-x86-64-linux-gnu`, `libcrypt-dev`, `libtsan2`, `gcc`, `libdpkg-perl`, `libubsan1`, `gcc-15`, `libfakeroot`, `lto-disabled-list`, and `gcc-15-x86-64-linux-gnu`, `libfile-fcntllock-perl`, and `make`. Suggested packages include `binutils-doc`, `autoconf`, `gcc-15-locales`, `debconf`, `autotools`, `gdb-x86-64-linux-gnu`, `debconf`, `libtool`, `git`, `g++-multilib`, `flex`, `bison`, `libstdc++-15-doc`, `gcc-15-doc`, `gcc-doc`, and `make-doc`. The summary shows that 31 packages will be installed, with a download size of 58.8 MB and a space needed of 205 MB. The terminal also shows the progress of downloading the packages from the Ubuntu archive.

```
abhihit-das@abhihit-das-VirtualBox: ~  
$ sudo apt install -y build-essential linux-headers-$(uname -r)  
linux-headers-7.0.0-31-generic is already the newest version (7.0.0-31.31).  
linux-headers-7.0.0-31-generic set to manually installed.  
The following packages were automatically installed and are no longer required:  
  linux-headers-7.0.0-30          linux-modules-7.0.0-30-generic  
  linux-headers-7.0.0-30-generic  linux-tools-7.0.0-30  
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30-generic  
  linux-main-modules-zfs-7.0.0-30-generic  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  build-essential  
  
Installing dependencies:  
  bzip2                gcc-x86-64-linux-gnu      libgcc-15-dev  
  dpkg-dev              libalgorithm-diff-perl   libhwasan0  
  fakeroot              libalgorithm-diff-xs-perl libitm1  
  g++                   libalgorithm-merge-perl liblsan0  
  g++-15                libasan8                  libquadmath0  
  g++-15-x86-64-linux-gnu libcc1-0                  libstdc++-15-dev  
  g++-x86-64-linux-gnu  libcrypt-dev              libtsan2  
  gcc                   libdpkg-perl              libubsan1  
  gcc-15                libfakeroot               lto-disabled-list  
  gcc-15-x86-64-linux-gnu libfile-fcntllock-perl   make  
  
Suggested packages:  
  binutils-doc          autoconf                  gcc-15-locales  
  debconf                autotools                 gdb-x86-64-linux-gnu  
  debconf                libtool                   git  
  g++-multilib           flex                      bison  
  g++-15-multilib        bison                    libstdc++-15-doc  
  gcc-15-doc              gcc-doc                   make-doc  
  gcc-multilib            gcc-15-multilib  
  
Summary:  
  Upgrading: 0, Installing: 31, Removing: 0, Not Upgrading: 22  
  Download size: 58.8 MB  
  Space needed: 205 MB / 38.5 GB available  
  
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libcc1-0 amd64 16-20260322-1ubuntu1 [51.2 kB]  
Get:2 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libitm1 amd64 16-20260322-1ubuntu1 [30.0 kB]  
Get:3 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libasan8 amd64 16-20260322-1ubuntu1 [3,184 kB]
```

Figure 7. 7. Installing Build Tools and Kernel Headers

8. Build Tools Installation Completed

Shows the build-essential installation completing successfully in the Ubuntu terminal.



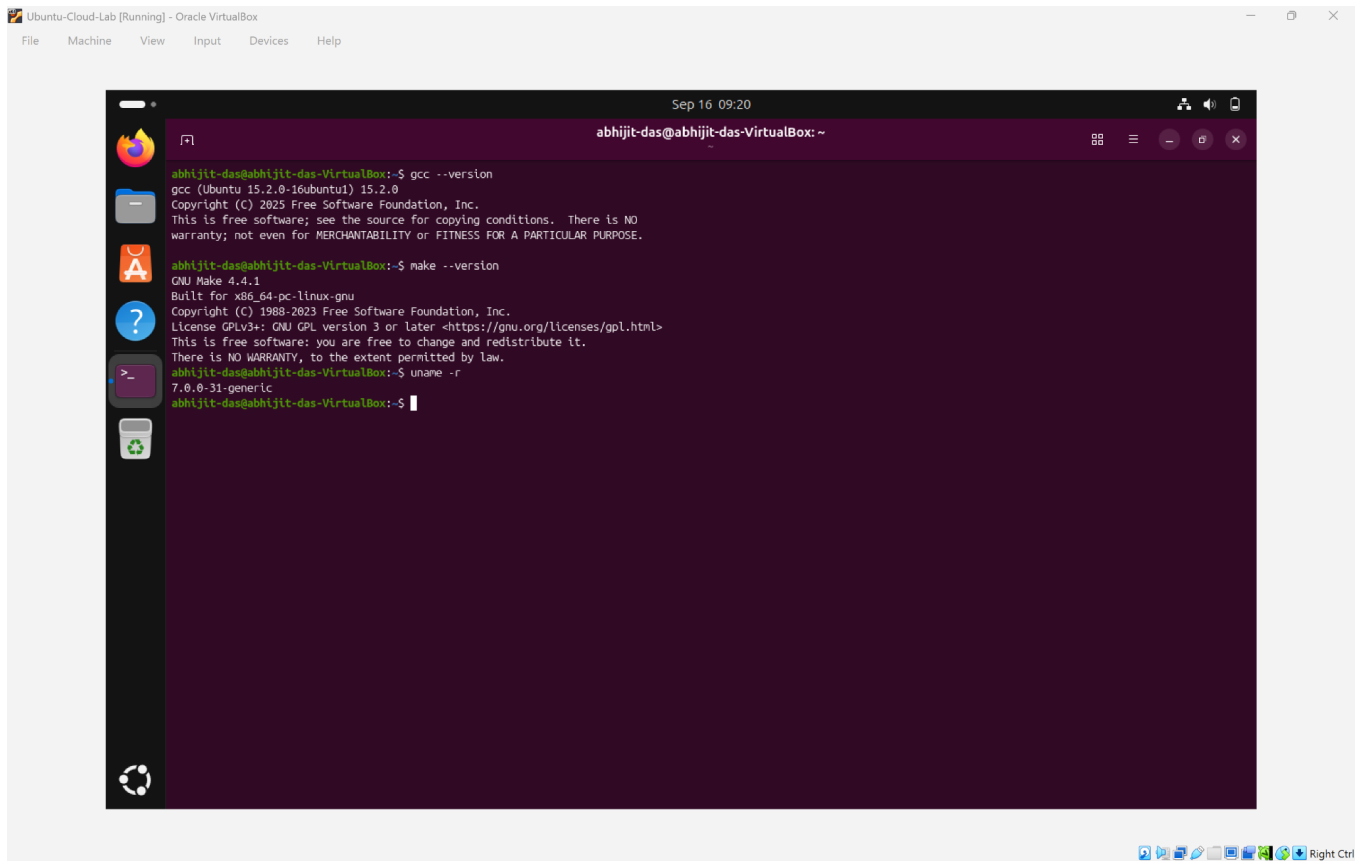
```

Sep 16 09:18
abhiit-das@abhiit-das-VirtualBox: ~
Preparing to unpack ./29-libalgorithm-merge-perl_0.08-5_all.deb...
Unpacking libalgorithm-merge-perl (0.08-5)...
Selecting previously unselected package libfile-fcntllock-perl.
Preparing to unpack ./30-libfile-fcntllock-perl_0.22-4ubuntu6_and64.deb...
Unpacking libfile-fcntllock-perl (0.22-4ubuntu6)...
Setting up lto-disabled-list (79)...
Setting up libfile-fcntllock-perl (0.22-4ubuntu6)...
Setting up libalgorithm-diff-perl (1.201-1)...
Setting up bzip2 (1.0.8-6ubuntu0.1)...
Setting up libfakeroot:amd64 (1.37.2-1)...
Setting up fakeroot (1.37.2-1)...
update-alternatives: using /usr/bin/fakeroot-sysv to provide /usr/bin/fakeroot (fakeroot) in auto mode
Setting up make (4.4.1-3)...
Setting up libquadmath0:amd64 (16-20260322-1ubuntu1)...
Setting up libdpkg-perl (1.23.7ubuntu1)...
Setting up libubsan1:amd64 (16-20260322-1ubuntu1)...
Setting up libhwasan0:amd64 (16-20260322-1ubuntu1)...
Setting up libcrypt-dev:amd64 (1:4.5.1-1)...
Setting up libasan8:amd64 (16-20260322-1ubuntu1)...
Setting up libtsan2:amd64 (16-20260322-1ubuntu1)...
Setting up libalgorithm-diff-xs-perl (0.04-9)...
Setting up libcc1-0:amd64 (16-20260322-1ubuntu1)...
Setting up liblsan0:amd64 (16-20260322-1ubuntu1)...
Setting up libitm1:amd64 (16-20260322-1ubuntu1)...
Setting up libalgorithm-merge-perl (0.08-5)...
Setting up libgcc-15-dev:amd64 (15.2.0-16ubuntu1)...
Setting up pkg-config (1.23.7ubuntu1)...
Setting up gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1)...
Setting up gcc-15 (15.2.0-16ubuntu1)...
Setting up libstdc++-15-dev:amd64 (15.2.0-16ubuntu1)...
Setting up g++-15-x86-64-linux-gnu (15.2.0-16ubuntu1)...
Setting up gcc-x86-64-linux-gnu (4:15.2.0-5ubuntu1)...
Setting up gcc (4:15.2.0-5ubuntu1)...
Setting up g++-x86-64-linux-gnu (4:15.2.0-5ubuntu1)...
Setting up g++-15 (15.2.0-16ubuntu1)...
Setting up g++ (4:15.2.0-5ubuntu1)...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.12ubuntu2.26.04.2)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2.4)...
abhiit-das@abhiit-das-VirtualBox: ~$
```

Figure 8. 8. Build Tools Installation Completed

9. Verification of Installed Tools and Kernel

Shows gcc --version, make --version and uname -r confirming the development tools and running kernel.



The screenshot shows a terminal window titled "abhihit-das@abhihit-das-VirtualBox: ~" with a dark purple background. The terminal displays the output of three commands: `gcc --version`, `make --version`, and `uname -r`. The `gcc` command output shows version 15.2.0. The `make` command output shows version 4.4.1. The `uname -r` command output shows the kernel version 7.0.0-31-generic. The terminal window is part of an Oracle VM VirtualBox interface, with a menu bar at the top (File, Machine, View, Input, Devices, Help) and a status bar at the bottom (Right Ctrl).

```
abhihit-das@abhihit-das-VirtualBox:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

abhihit-das@abhihit-das-VirtualBox:~$ make --version
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
abhihit-das@abhihit-das-VirtualBox:~$ uname -r
7.0.0-31-generic
abhihit-das@abhihit-das-VirtualBox:~$
```

Figure 9. 9. Verification of Installed Tools and Kernel

Result and Conclusion

The Oracle VirtualBox environment was configured and the Ubuntu virtual machine was successfully booted. The VM configuration, storage, and network settings were verified. Ubuntu package information was updated, build-essential and the required Linux kernel headers were installed, and the environment was verified using GCC, GNU Make, and the kernel version command.

Commands demonstrated

- `sudo apt update`
- `sudo apt install -y build-essential linux-headers-$(uname -r)`
- `gcc --version`
- `make --version`
- `uname -r`